Branch Coverage Report — Online Examination Portal

Course: UE24CS341A — Software Engineering, PES University, Dept. of CSE
Project: Online Examination System with Multiple-Choice Questions and Evaluation
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Build measured: commit `2d74453` · **Baseline:** `Documents/Test_Plan.pdf` v4.0

1. Summary

The automated test suite takes **72.1% of the application's branches** (199 of 276). That is **76.5%** (199 of 260) without a developer seeding command that no test runs.

The decisions that matter most are fully covered: every branch in the marking engine (`evaluation`), in results publication (`results`), and in the role check that guards every page (`identity/access.py`). Most of the 77 untaken branches are the "no" side of a defensive check that no test triggers, such as a request about an archived course or malformed JSON sent to the question API. Twenty-one of them, though, belong to behaviour that has no test at all, and §5 lists them as the follow-up work.

Statement coverage for the same run is in `Documents/Code_Coverage_Report.pdf`.

2. What branch coverage measures

Every `if`, loop or other decision can go two or more ways, and each way is a branch. Coverage.py records which ones the tests actually took. Branch coverage is stricter than statement coverage: an `if` without an `else` can have every statement run while the "condition false" path is never tried. Of the 276 branches, 45 sit on a **partial** decision, where the tests took one way and never the other.

Tool	Coverage.py 7.16.2 with branch measurement on (`branch = True` in `.coveragerc`)
Command	`coverage run manage.py test -v 2`, then `coverage report` and `coverage json`
Tests run	227: all passed, none skipped (the same run as the Code Coverage Report and the Test Plan results)
Not measured	Migrations, the test files, `manage.py`, the WSGI/ASGI entry points, `loadtest/`

3. Overall result

Measure	Taken	Total	Coverage
Branches, whole application	199	276	**72.1%**
Branches, without the seed command	199	260	**76.5%**
Decisions only partly covered		45	

4. Branch coverage by module

Module	What it holds	Branches taken	Coverage
`evaluation`	Marking engine: FR-022, FR-023, FR-024	8 / 8	**100%**
`results`	Withholding and publishing results, scorecard: FR-025 to FR-027	10 / 10	**100%**
`attempts`	Starting, answering, autosave, resume, auto-submission: FR-015 to FR-021	44 / 48	91.7%

Module	What it holds	Branches taken	Coverage
`questionbank`	Question authoring, tagging, versioning: FR-007 to FR-009	36 / 48	75.0%
`examsetup`	Building, publishing and scheduling examinations: FR-010 to FR-014	44 / 60	73.3%
`integrity`	Focus-loss events and audit-log viewer: FR-028 to FR-030	4 / 6	66.7%
`identity`	Registration, sign-in, roles, courses, enrolment: FR-001 to FR-006	44 / 68	64.7%
`repository`	Data model, audit writer, course form, seed command	9 / 28	32.1%
`repository` without the seed command		9 / 12	75.0%
`portal`	Settings and routing; no decisions	0 / 0	n/a
Total		**199 / 276**	**72.1%**

Inside `identity`, the access-control module that enforces FR-003 on every request (`identity/access.py`) is at 8 of 8. The low figure for the app as a whole comes from the course administration views (§5).

5. The 77 untaken branches

The untaken branches fall into six groups. Only the first group is missing behaviour; the others are paths the existing tests did not need to take.

Group	Branches	What they are	Assessment
Behaviour with no test	21	Administrator course editing, including the duplicate-code check on edit (11). Unenrolling a Student and the "not your course" refusal (5). The dashboard placing an examination with no window, or a missed one whose window ended unsubmitted (2). Refusing to delete a question in a published examination, and restoring the previous version on delete (2). Ending a live session of a Student whose data is erased under NFR-008 (1)	**Follow-up.** Each needs a test keyed to a Test Plan ID
Refusals never triggered	14	Acting on an archived course (4). Changing an examination no longer in Draft (2). Zero or negative marks (2). A duplicate or missing question in a paper (2). Mismatched passwords (1). A duplicate caught by the database during concurrent registration (2). Archiving an already-archived course (1)	The rule exists in code but no test asserts it. The marks and archived-course refusals are the most worth adding
Malformed input	9	The question API receiving a non-object body, non-list options or a non-object option, reporting invalid options back, or getting a GET (6). A non-numeric examination ID (1). A non-numeric ID in the audit-log filter (1). An anonymous actor passed to the audit writer (1)	Hardening paths. Low risk, since none is reachable through the portal's own pages
Plain GETs and no-op paths	12	Showing an empty form (6). Re-showing a form after an invalid POST (2). Saving an examination with no field changed (1). The post-sign-in `next` redirect (1). Opening a question page of an already-submitted attempt (1). The audit log opened with no filter (1)	Low risk. The POSTs these sit beside are tested
Form internals	5	A form built without a user, a form saved with `commit=False`, and early returns when other fields already failed	No user-visible effect on their own
Seed command	16	`seed_bank`, the developer command that fills a demo question bank	Not application behaviour. No test needed

6. Conclusion

Branch coverage is 72.1% overall and 76.5% without the seed command. The decision points that decide a Student's marks, whether a result is visible, and who may open a page are all covered.

The 21 branches of untested behaviour in §5 are the follow-up list for the optional Sprint 3. Closing them, together with the marks and archived-course refusals, would add 27 branches and lift branch coverage to about 87% without the seed command.